

Welcome!

Nameplates please. And technology encouraged today!

All TF materials are available at github.com/nolankav/api-203.

If you want to follow along, download the dataset here:

In R: `df <- read.csv("http://tinyurl.com/api-203-tf-4")`

Parks and Prediction

API 203: TF Session 4

R

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April 4, 2025



Goals for today

- 1. Briefly touch on the principles of non-linear models.**
- 2. Contrast the goals of causal inference and prediction.**
- 3. Review a few strategies for predicting modeling, including LASSO.**
- 4. Learn how to generate predictive models in R.**
- 5. Practice evaluating the accuracy of predictive models.**

We'll treat this session like a workshop with an interactive example.

Overview of our sample data

Dataset of ~17,000 U.S. adults surveyed in 2019

casein	Respondent identifier	Cooperative Election Study
commonweight	Post-stratification survey weight	Cooperative Election Study
age	Respondent age (in years)	Cooperative Election Study
gender	Respondent gender	Cooperative Election Study
race_eth	Respondent race/ethnicity	Cooperative Election Study
education	Respondent educational attainment	Cooperative Election Study
marital	Respondent marital status	Cooperative Election Study
party_scale	Party identification (-3 = strong Rep. to 3 = strong Dem.)	Cooperative Election Study
public_ins	Respondent has public health insurance (1) or not (0)	Cooperative Election Study
public_option	Supports a public option in Medicare (1) or not (0)	Cooperative Election Study

A brief detour on non-linear models

- **Linear models are a good default, absent evidence of non-linearity.**
- **But when we want a non-linear model, we have several options.**
 - Higher-order polynomials: $Y = \beta_0 + \beta_1 * X + \beta_2 * X^2 + \dots$
 - Transformations to Y and/or X, such as logs: $\ln(Y) = \beta_0 + \beta_1 * X + \dots$
 - Others: percentiles/ranks, Winsorization
- **With non-linear models, we must be careful about interpretation.**

A brief detour on non-linear models

Model	Regression Equation (PRF)	Interpretation of β_1 (Math)	Interpretation of β_1 (Words)
Level-Level	$Y = \beta_0 + \beta_1(X_1) + u$	$\Delta Y = (\beta_1)\Delta X_1$	A 1 unit increase in X_1 is associated with a β_1 change in Y
Level-Log	$Y = \beta_0 + \beta_1(\ln X_1) + u$	$\Delta Y = (\frac{\beta_1}{100})\% \Delta X_1$	A 1% increase in X_1 is associated with a $0.01\beta_1$ change in Y
Log-Level	$\ln(Y) = \beta_0 + \beta_1(X_1) + u$	$\% \Delta Y = (100\beta_1)\Delta X_1$	A 1 unit increase in X_1 is associated with a $100\beta_1\%$ change in Y
Log-Log	$\ln(Y) = \beta_0 + \beta_1(\ln X_1) + u$	$\% \Delta Y = (\beta_1)\% \Delta X_1$	A 1% increase in X_1 is associated with a $\beta_1\%$ change in Y

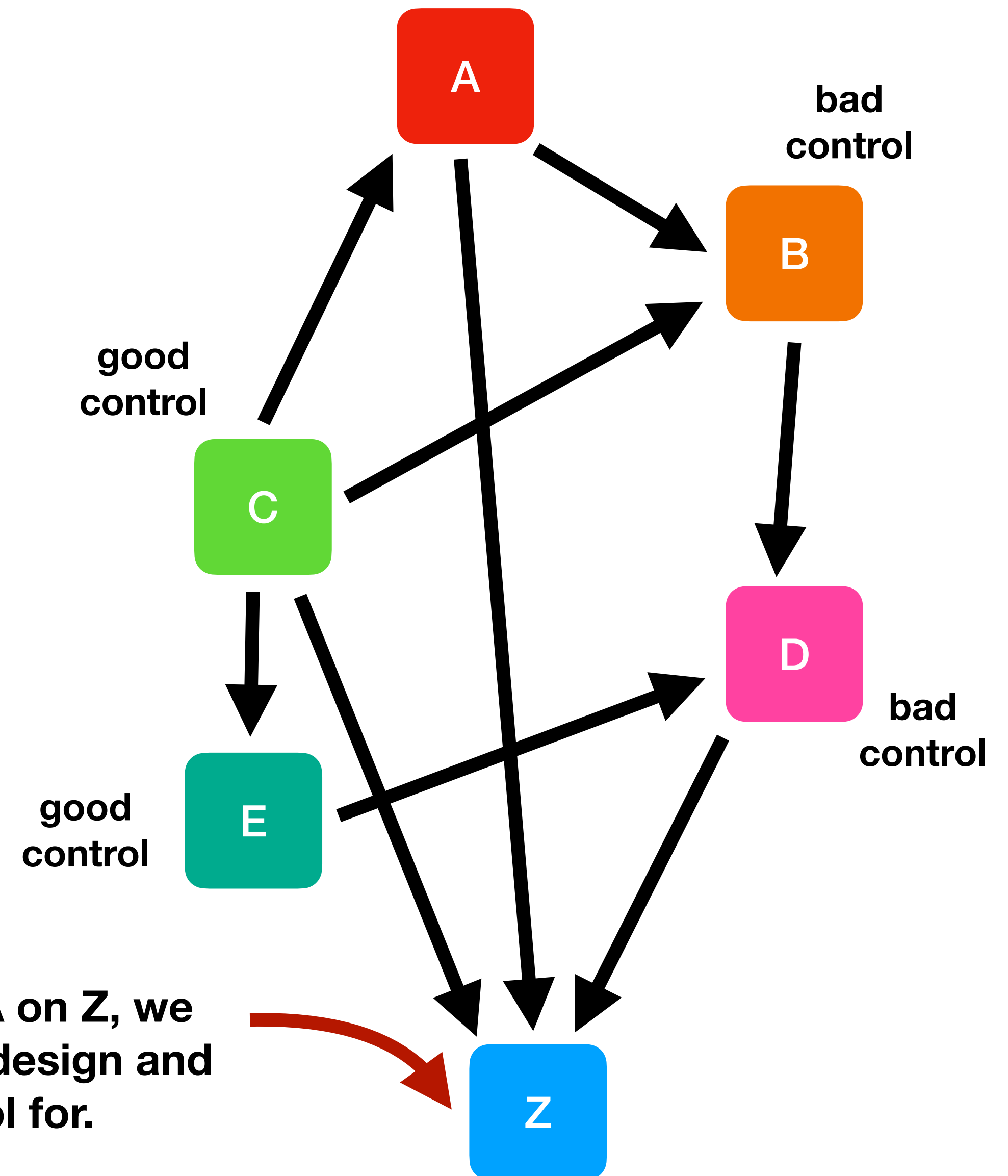
Finally, the “percent” interpretation we are often tempted to make with non-transformed linear models!

What's our goal?

With causal inference, the name of the game was bias: what causes it and how to get rid of it.

Why? Our goal was to estimate an unbiased effect.

To get the causal estimate of A on Z, we have to be careful about study design and which variables we control for.



What's our goal?

With causal inference, the name of the game was bias: what causes it and how to get rid of it.

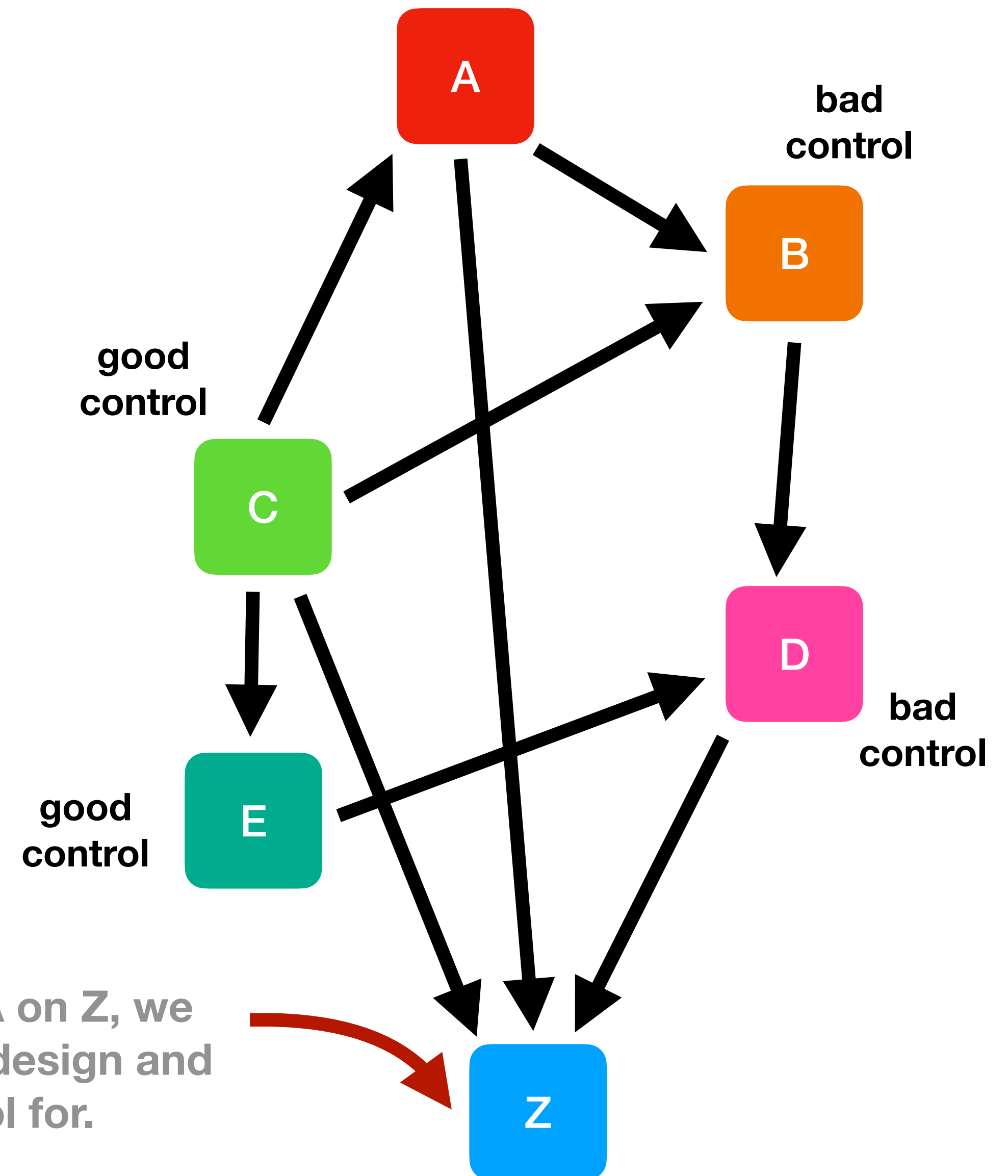
Why? Our goal was to estimate an unbiased effect.

With prediction, we don't care about bias. In fact, we invite bias. We'll use whatever we can.

Why? Our goal is accurate prediction.

To get the causal estimate of A on Z, we have to be careful about study design and which variables we control for.

But if we just want to predict Z, every relevant variable is fair game.



What's our goal?

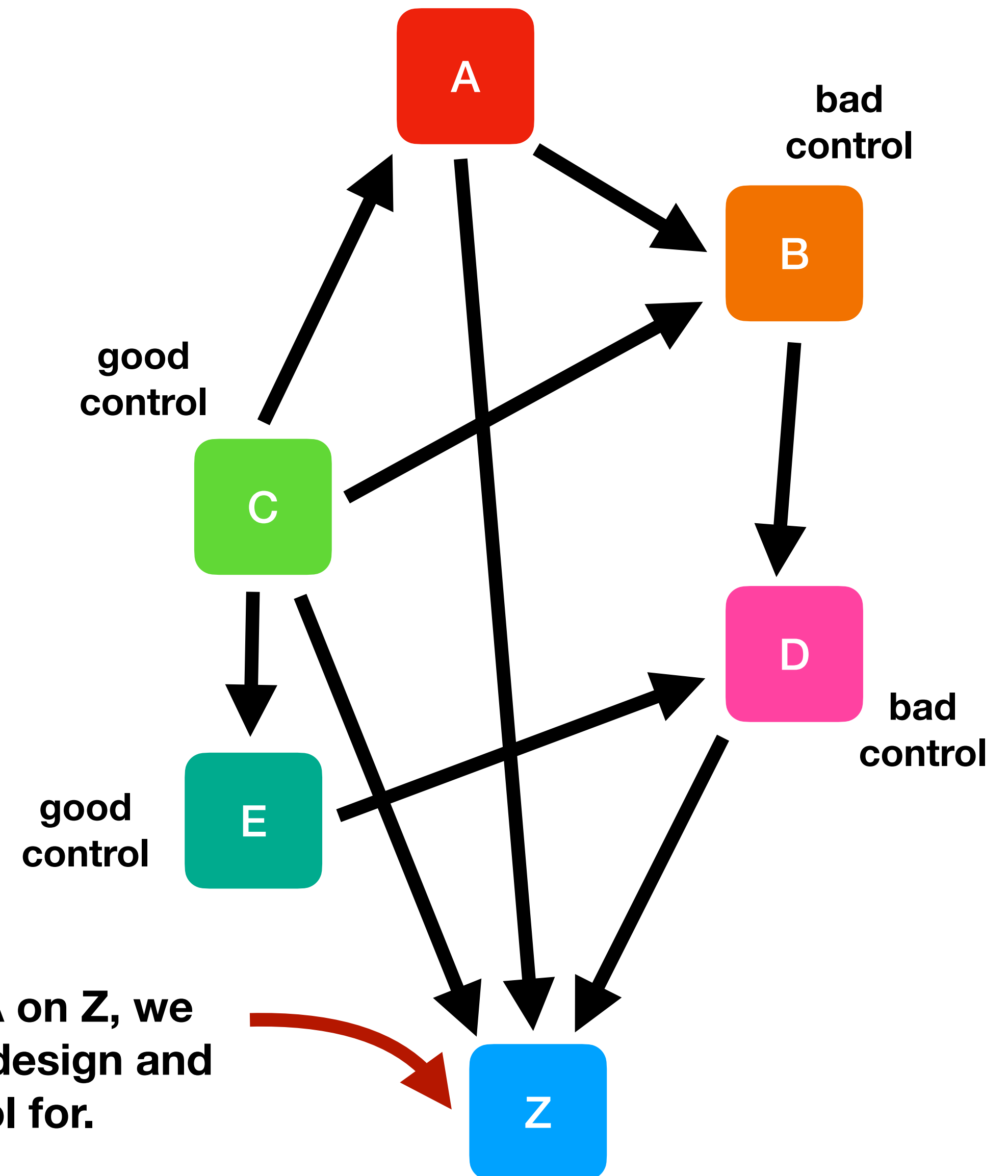
With all possible variables at our disposal, we do have to worry about overfitting our models.

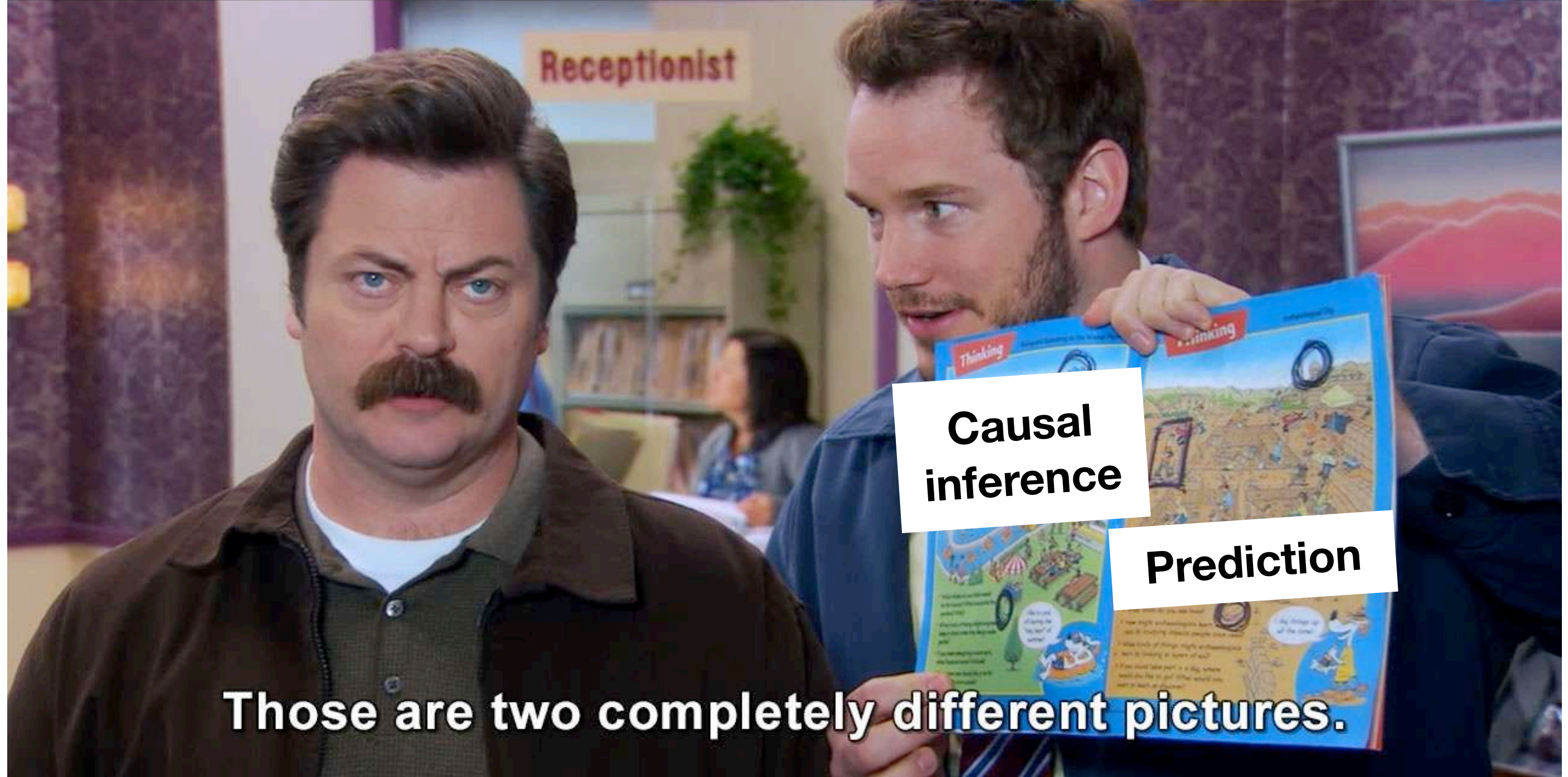
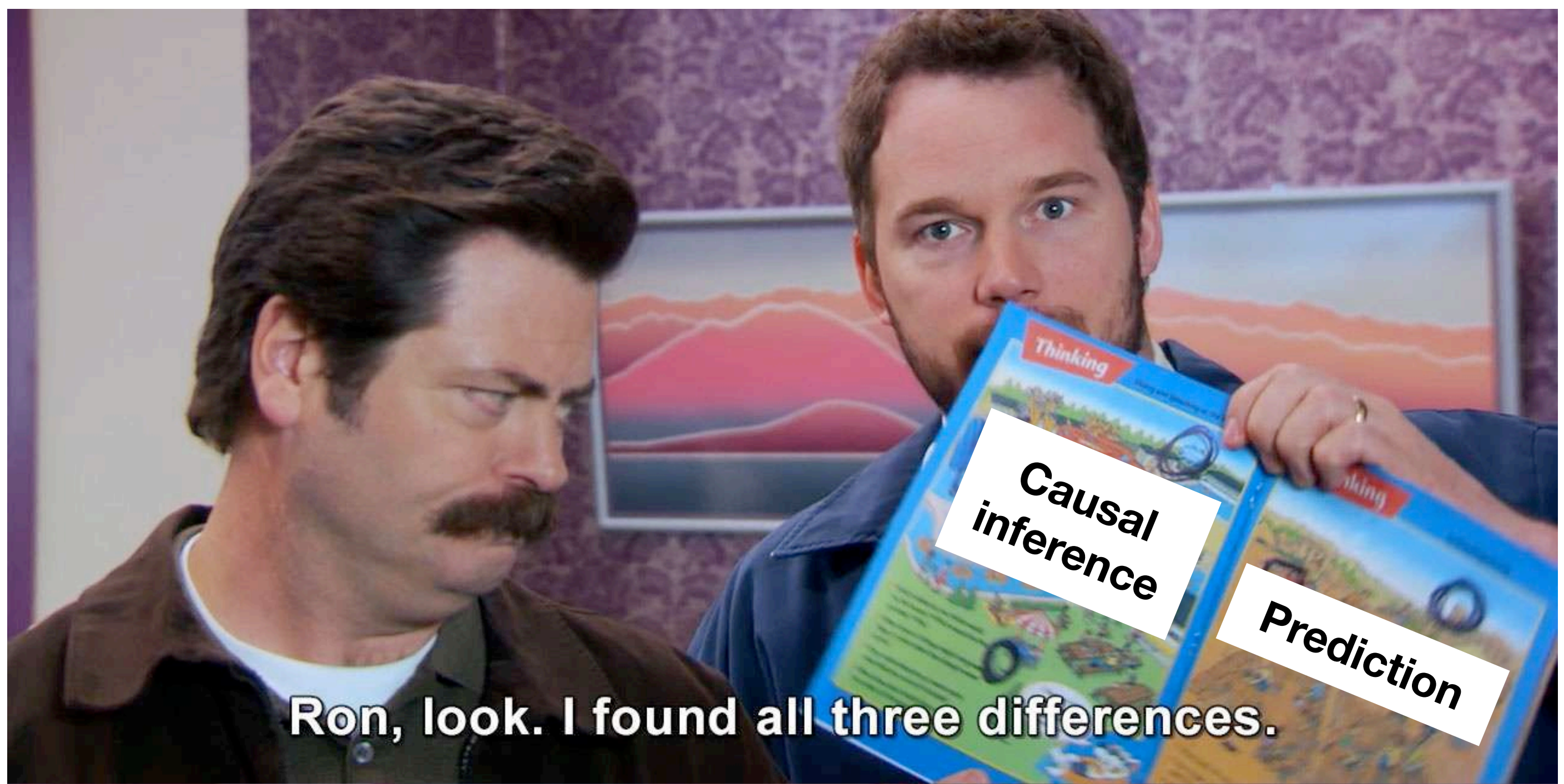
Overfitted models look good in-sample, but they tend to perform poorly out-of-sample.

Above all, we want useful predictions!

To get the causal estimate of A on Z, we have to be careful about study design and which variables we control for.

But if we just want to predict Z, every relevant variable is fair game.





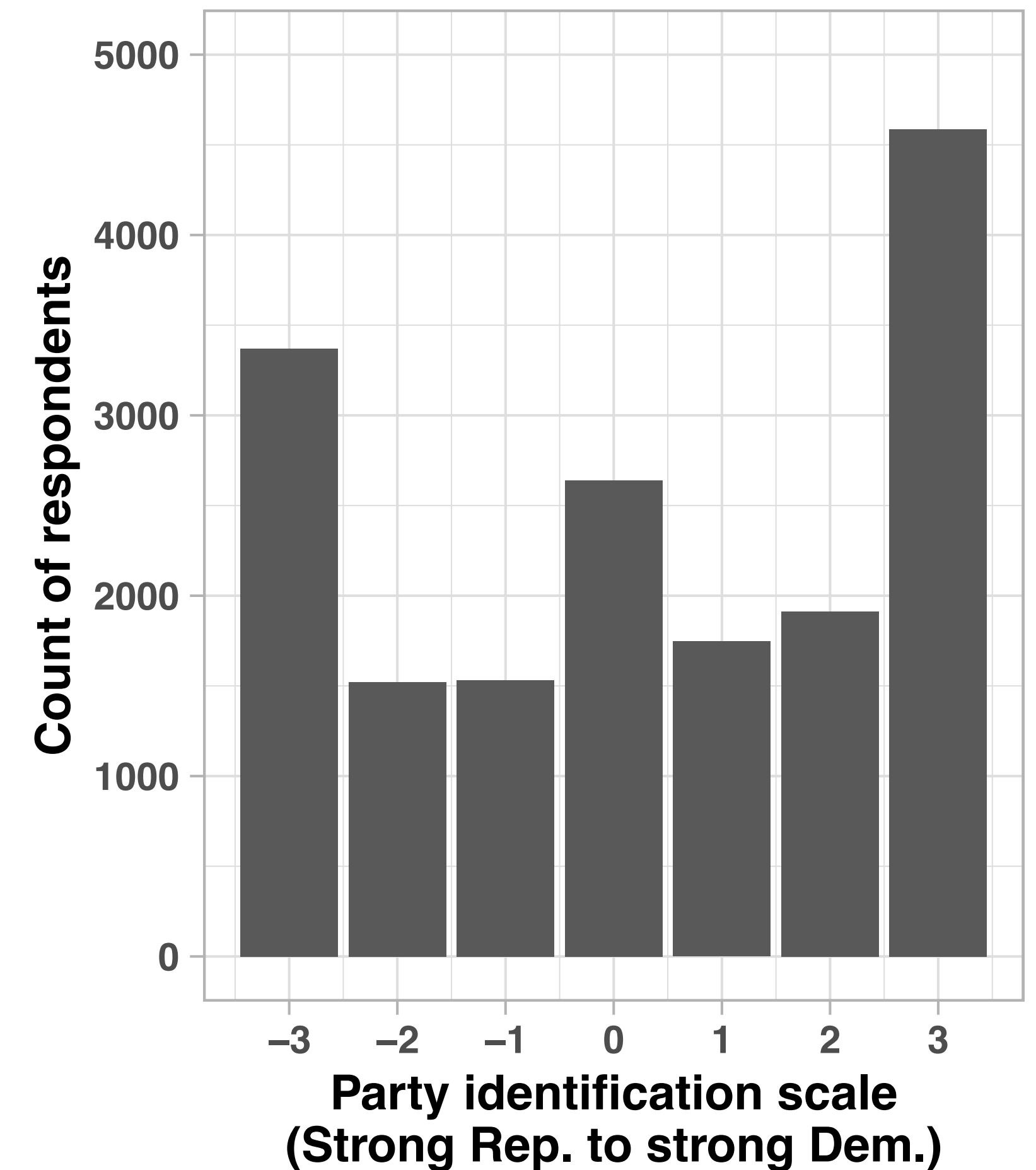
OK, so what are we predicting?

Let's try to predict partisan strength.

Parties want to know whom they can contact for fundraising, get out the vote, etc.

This is “supervised” machine learning since we can validate our predictions on the “truth”.

```
# Generate bar plot of parties
plot_1 <- ggplot(data=df, aes(x=party_scale)) +
  geom_bar(stat="count") +
  xlab("Party identification scale\n(Strong Rep. to strong Dem.)") +
  ylab("Count of respondents") +
  theme_light() +
  theme(text = element_text(size = 10, face = "bold")) +
  scale_x_continuous(breaks = seq(-3, 3, 1)) +
  scale_y_continuous(limits = c(0,5000),
                     breaks = seq(0,5000,1000))
```



First, let's split our data into two.

We want to designate “training” and “test” sets. One is for developing our model, and the other is for validating its predictions.

Here, we will do a 70/30 split. Generally, we want more data in the training set.

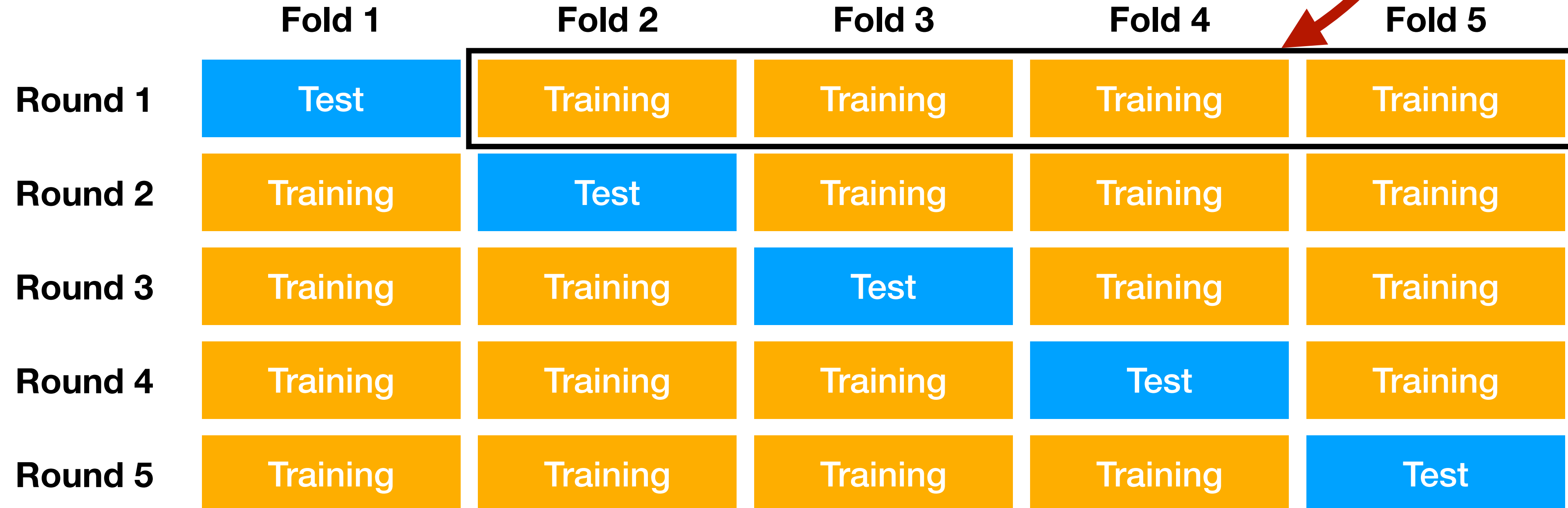
```
# Designate training and test sets
set.seed(1234)
split      <- initial_split(df, 0.7)
df_train   <- training(split)
df_test    <- testing(split)
```

An advanced approach is K-fold cross-validation.

This approach splits the data into K equal folds.

Then, we use each fold as the test set for one round of modeling and average across all rounds.

In each round, we pool all the training folds into one big training set.



	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Round 1	Test	Training	Training	Training	Training
Round 2	Training	Test	Training	Training	Training
Round 3	Training	Training	Test	Training	Training
Round 4	Training	Training	Training	Test	Training
Round 5	Training	Training	Training	Training	Test

A photograph of a man and a woman sitting at a wooden table, each holding a glass of red wine. The man, on the left, is wearing a dark suit and tie, looking towards the woman. The woman, on the right, is wearing a grey and white striped sweater with a colorful orange and yellow scarf, smiling at the man. The background shows a wooden door and a glimpse of another room with a white table and chairs. Three white text boxes with black text are overlaid on the image.

Predictive modeling

**Ruining springtime
for MPPs**

Causal inference

Let's train a very basic model.

```
# OLS regression
```

```
model_1 <- lm(party_scale ~ gender, data=df_train)
summary(model_1)
```

Call:

```
lm(formula = party_scale ~ gender, data = df_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4210	-2.0718	-0.0718	2.5790	2.9282

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.07180	0.03052	2.353	0.0186 *
genderFemale	0.34924	0.04092	8.535	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.238 on 12112 degrees of freedom

Multiple R-squared: 0.005979, Adjusted R-squared: 0.005897

F-statistic: 72.85 on 1 and 12112 DF, p-value: < 2.2e-16

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(Intercept)   0.07180    0.03052   2.353   0.0186 *
genderFemale   0.34924    0.04092   8.535  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.238 on 12112 degrees of freedom
Multiple R-squared:  0.005979,    Adjusted R-squared:  0.005897
F-statistic: 72.85 on 1 and 12112 DF,  p-value: < 2.2e-16
```

One way of evaluating predictive models is R-squared. It's less ideal than root mean square error (RMSE), which is measured in units of our outcome, but R-squared can be useful at a glance.

Here, the R-squared is low, so maybe we should temper our expectations...

Multiple R-squared: 0.005979, Adjusted R-squared: 0.005897

Let's evaluate our predictions!

```
# In-sample prediction & MSE
```

```
df_train$predict <- predict(model_1, df_train)
```

```
df_train %>% summarise(mse = mean((party_scale - predict)^2))
```

```
mse
```

```
1 5.005678
```

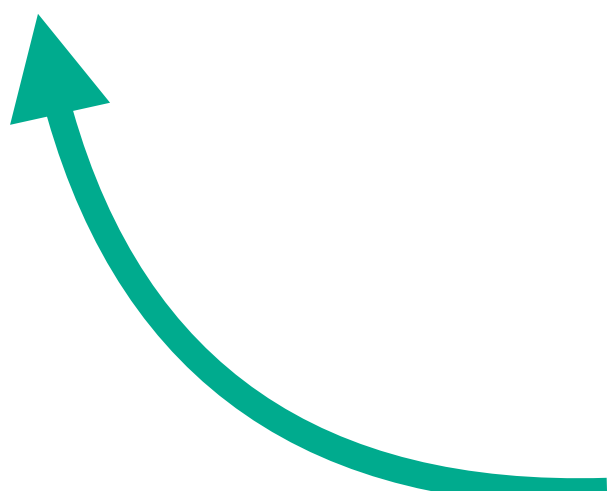
```
# Out-of-sample prediction & MSE
```

```
df_test$predict <- predict(model_1, df_test)
```

```
df_test %>% summarise(mse = mean((party_scale - predict)^2))
```

```
mse
```

```
1 5.016616
```



To compute the MSE, we take the error between the true and predicted values, square them, and take the mean.

Hence, mean squared error.

Let's evaluate our predictions!

```
# In-sample prediction & MSE
```

```
df_train$predict <- predict(model_1, df_train)
df_train %>% summarise(mse = mean((party_scale - predict)^2))
```

```
      mse
1 5.005678
```

```
# Out-of-sample prediction & MSE
```

```
df_test$predict <- predict(model_1, df_test)
df_test %>% summarise(mse = mean((party_scale - predict)^2))
```

```
      mse
1 5.016616
```

To get the RMSE, we just take the square root:

$\text{sqr}(5.01) = 2.24$

$\text{sqr}(5.02) = 2.24$

Remember that our scale is -3 to 3 , so on average, we are off by a fair amount!

Also, notice that the in-sample predictions are a tiny bit better than out-of-sample predictions.

Let's train a more complicated model.

```
# OLS regression
model_2 <- lm(party_scale ~ age + gender + race_eth + education, data=df_train)
summary(model_2)
```

```
Call:
lm(formula = party_scale ~ age + gender + race_eth + education,
    data = df_train)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-5.8095 -1.9455  0.1316  1.7723  3.9824
```

```
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.479182   0.133753   11.059 < 2e-16 ***
age             -0.014695   0.001157  -12.696 < 2e-16 ***
genderFemale      0.232517   0.038988    5.964 2.53e-09 ***
race_ethBlack     1.406425   0.138982   10.119 < 2e-16 ***
race_ethHispanic/Latino 0.146127   0.137354    1.064  0.2874
race_ethNative American -0.429021   0.216037   -1.986  0.0471 *
race_ethOther     -0.245887   0.164220   -1.497  0.1343
race_ethWhite     -0.519662   0.127776   -4.067 4.79e-05 ***
educationHigh school graduate -0.722277   0.051289  -14.082 < 2e-16 ***
educationLess than high school -0.539031   0.093625   -5.757 8.75e-09 ***
educationSome college or 2-year degree -0.518299   0.047272  -10.964 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.119 on 12103 degrees of freedom
Multiple R-squared:  0.1092, Adjusted R-squared:  0.1085
F-statistic: 148.3 on 10 and 12103 DF,  p-value: < 2.2e-16
```

Let's train a more complicated model.

```
# OLS regression
model_2 <- lm(party_scale ~ age + gender + race_eth + education, data=df_train)
summary(model_2)
```

```
Call:
lm(formula = party_scale ~ age + gender + race_eth + education,
    data = df_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.8095	-1.9455	0.1316	1.7723	3.9824

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.479182	0.133753	11.059	< 2e-16	***
age	-0.014695	0.001157	-12.696	< 2e-16	***
genderFemale	0.232517	0.038988	5.964	2.53e-09	***
race_ethBlack	1.406425	0.138982	10.119	< 2e-16	***
race_ethHispanic/Latino	0.146127	0.137354	1.064	0.2874	
race_ethNative American	-0.429021	0.216037	-1.986	0.0471	*
race_ethOther	-0.245887	0.164220	-1.497	0.1343	
race_ethWhite	-0.519662	0.127776	-4.067	4.79e-05	***
educationHigh school graduate	-0.722277	0.051289	-14.082	< 2e-16	***
educationLess than high school	-0.539031	0.093625	-5.757	8.75e-09	***
educationSome college or 2-year degree	-0.518299	0.047272	-10.964	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.119 on 12103 degrees of freedom

Multiple R-squared: 0.1092, Adjusted R-squared: 0.1085

F-statistic: 148.3 on 10 and 12103 DF, p-value: < 2.2e-16

**The R-squared went up: 0.01 vs. 0.11.
That's a good sign!**



Let's evaluate our predictions!

```
# In-sample prediction & MSE
```

```
df_train$predict_2 <- predict(model_2, df_train)
```

```
df_train %>% summarise(mse = mean((party_scale - predict_2)^2))
```

```
      mse
```

```
1 4.485947
```

```
# Out-of-sample prediction & MSE
```

```
df_test$predict_2 <- predict(model_2, df_test)
```

```
df_test %>% summarise(mse = mean((party_scale - predict_2)^2))
```

```
      mse
```

```
1 4.553173
```

Let's evaluate our predictions!

```
# In-sample prediction & MSE
```

```
df_train$predict_2 <- predict(model_2, df_train)
```

```
df_train %>% summarise(mse = mean((party_scale - predict_2)^2))
```

```
mse
```

```
1 4.485947
```

```
# Out-of-sample prediction & MSE
```

```
df_test$predict_2 <- predict(model_2, df_test)
```

```
df_test %>% summarise(mse = mean((party_scale - predict_2)^2))
```

```
mse
```

```
1 4.553173
```

To get the RMSE, we take the square root:

$\text{sqr}(4.49) = 2.12$

$\text{sqr}(4.55) = 2.13$

This is better than our last model!



**Making predictions with
random models.**

“I have no idea what I’m doing,”



**Making predictions with
random models.**

“But I know I’m doing it well”

Let's bring out the big guns.

We'll use every variable and interact them!

```
# OLS regression
model_3 <- lm(party_scale ~ age*gender*race_eth*education*
marital*public_ins*public_option, data=df_train)

# In-sample prediction & MSE
df_train$predict_3 <- predict(model_3, df_train)
df_train %>% summarise(mse = mean((party_scale - predict_3)^2))

      mse
1 3.585402

# Out-of-sample prediction & MSE
df_test$predict_3 <- predict(model_3, df_test)
df_test %>% summarise(mse = mean((party_scale - predict_3)^2))

      mse
1 252.4782
```

...including some variables, like political opinions, that are causally downstream of party affiliation!

But that's OK: Our goal is predictive accuracy, not causal identification.

Let's bring out the big guns.

We'll use every variable and interact them!

```
# OLS regression
model_3 <- lm(party_scale ~ age*gender*race_eth*education*
marital*public_ins*public_option, data=df_train)

# In-sample prediction & MSE
df_train$predict_3 <- predict(model_3, df_train)
df_train %>% summarise(mse = mean((party_scale - predict_3)^2))
```

```
      mse
1 3.585402
```

```
# Out-of-sample prediction & MSE
df_test$predict_3 <- predict(model_3, df_test)
df_test %>% summarise(mse = mean((party_scale - predict_3)^2))
```

```
      mse
1 252.4782
```

Our in-sample predictions got better...
 $\text{sqr}(3.59) = 1.89$

...but out-of-sample, they're terrible!
 $\text{sqr}(252.48) = 15.89$

This is what happens when we “overfit” our model.
We have only 12,114 people in the training set yet fit 1,920 coefficients!

We need some guardrails.

Enter: LASSO regression.

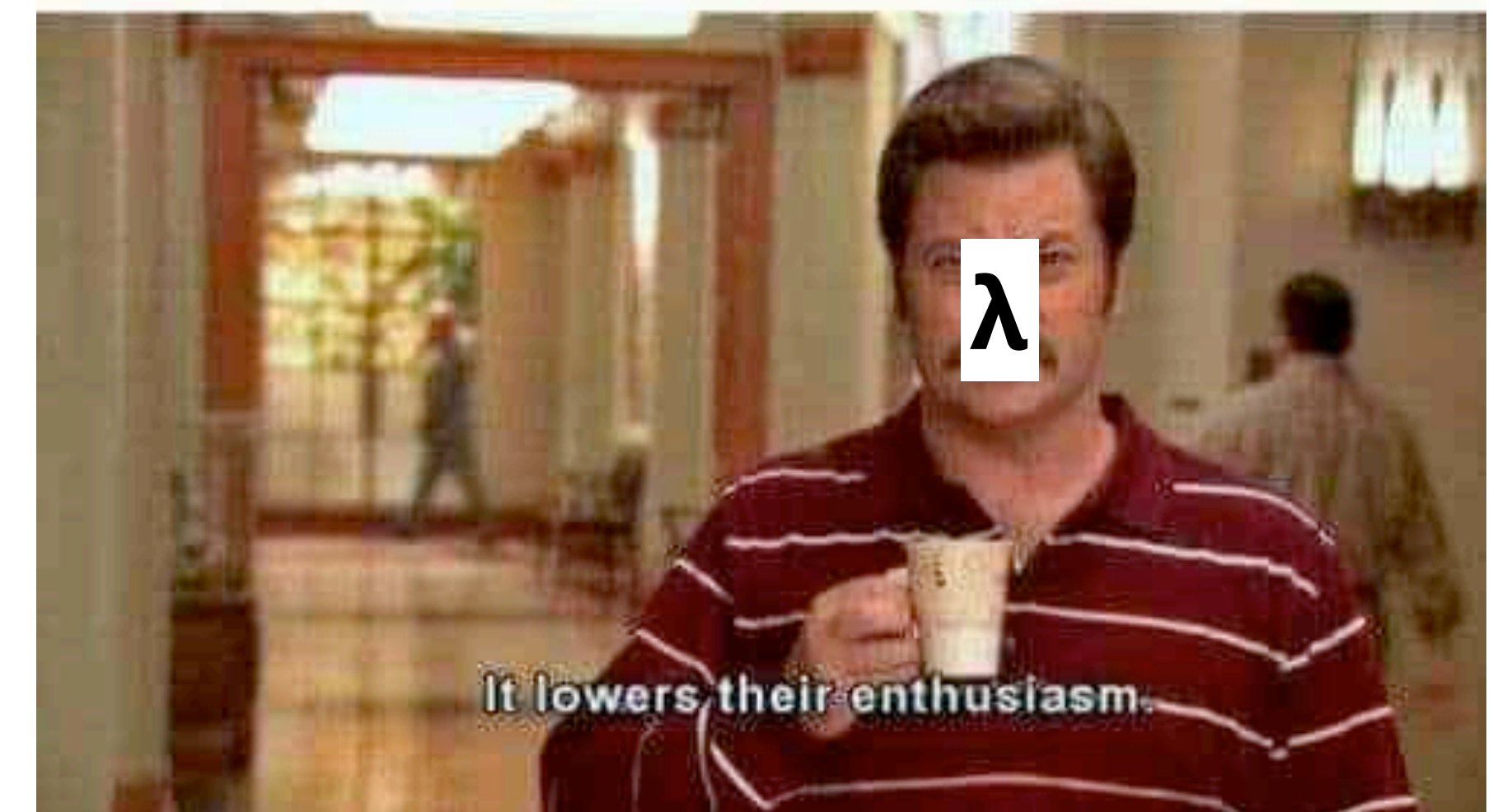
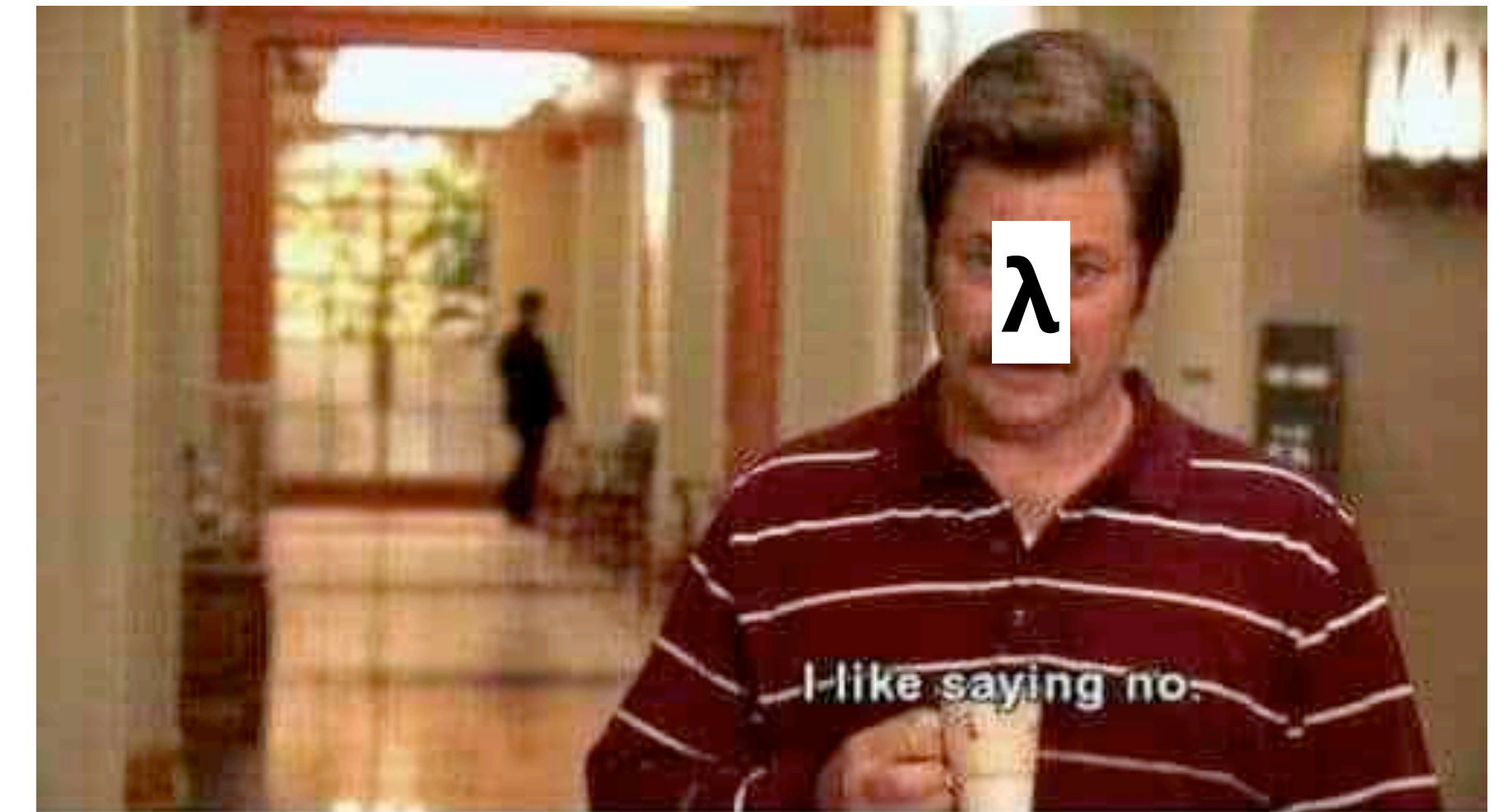
This is normal OLS.

$$\hat{\beta}_{\lambda}^{LASSO} = \arg \min_{\beta} \left[\underbrace{\sum_{i=1}^n \left(y_i - (\beta_0 + \sum_{j=1}^p \beta_j x_{ij}) \right)^2}_{\text{Loss function}} + \underbrace{\lambda \sum_{j=1}^p |\beta_j|}_{\text{Regularizer}} \right]$$

Where $\lambda \geq 0$ is called a *tuning parameter*.

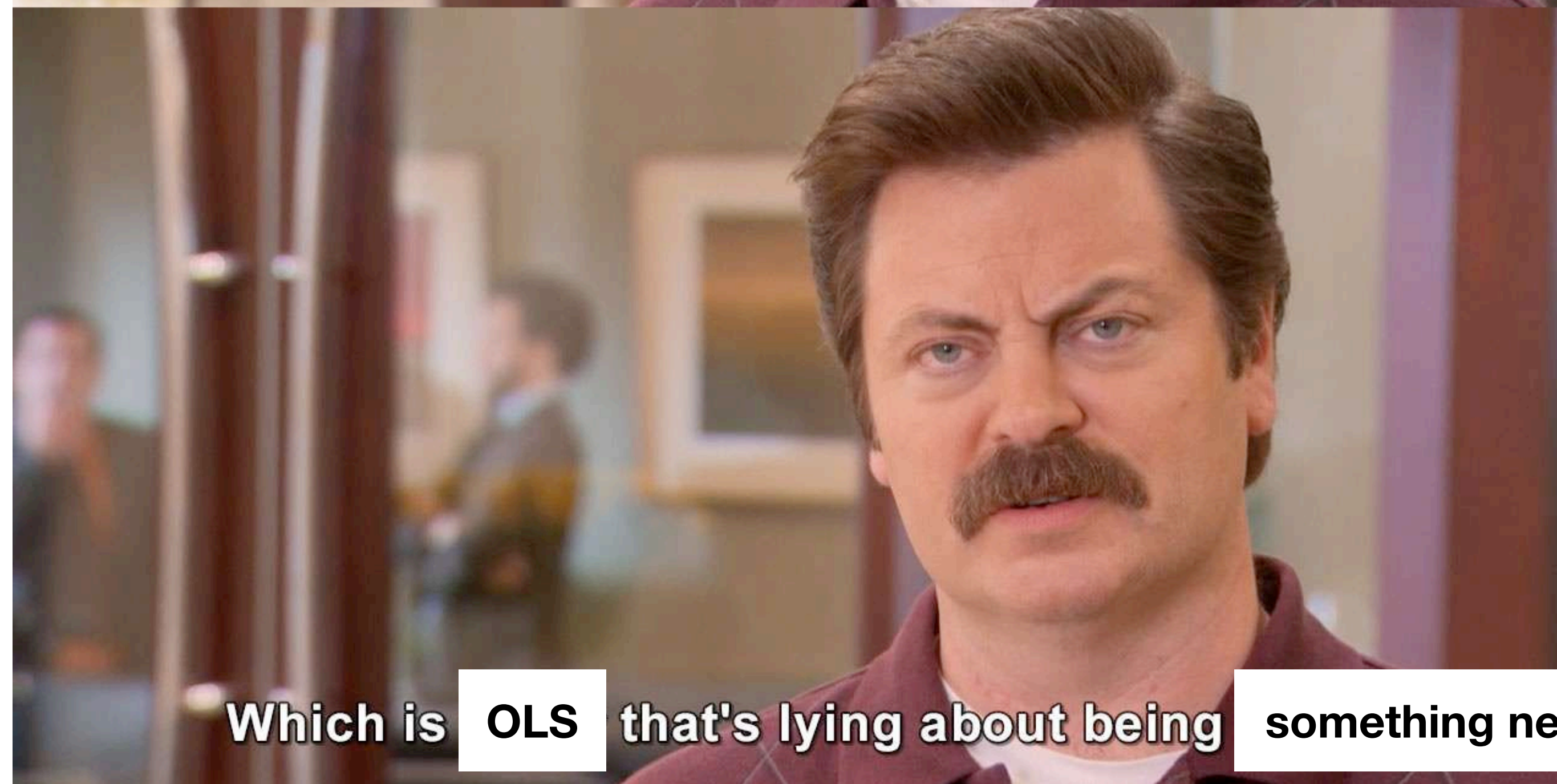
This penalizes us for more and larger betas.

This way, we reach a point at which adding more betas doesn't improve fit.





There's only one thing I hate more than **OLS**, **LASSO** .



Which is **OLS** that's lying about being something new and fancy

Alright, here we go.

First, we will load the glmnet library.

```
# Load library  
library(glmnet)
```

Second, we will select the Y and X variables that we want to use.

```
# Set the variables that are "fair game"  
Y <- df_train$party_scale  
X <- data.matrix(df_train[, c("age", "gender", "race_eth", "education",  
                             "marital", "public_ins", "public_option")])
```

Third, we run the model.

```
# LASSO regression  
lasso <- cv.glmnet(x=X, y=Y)  
plot(lasso); log(lasso$lambda.min)
```

```
[1] -6.125561
```

Lastly, we will evaluate the lambda that was selected.

Alright, here we go.

```
# Load library
library(glmnet)

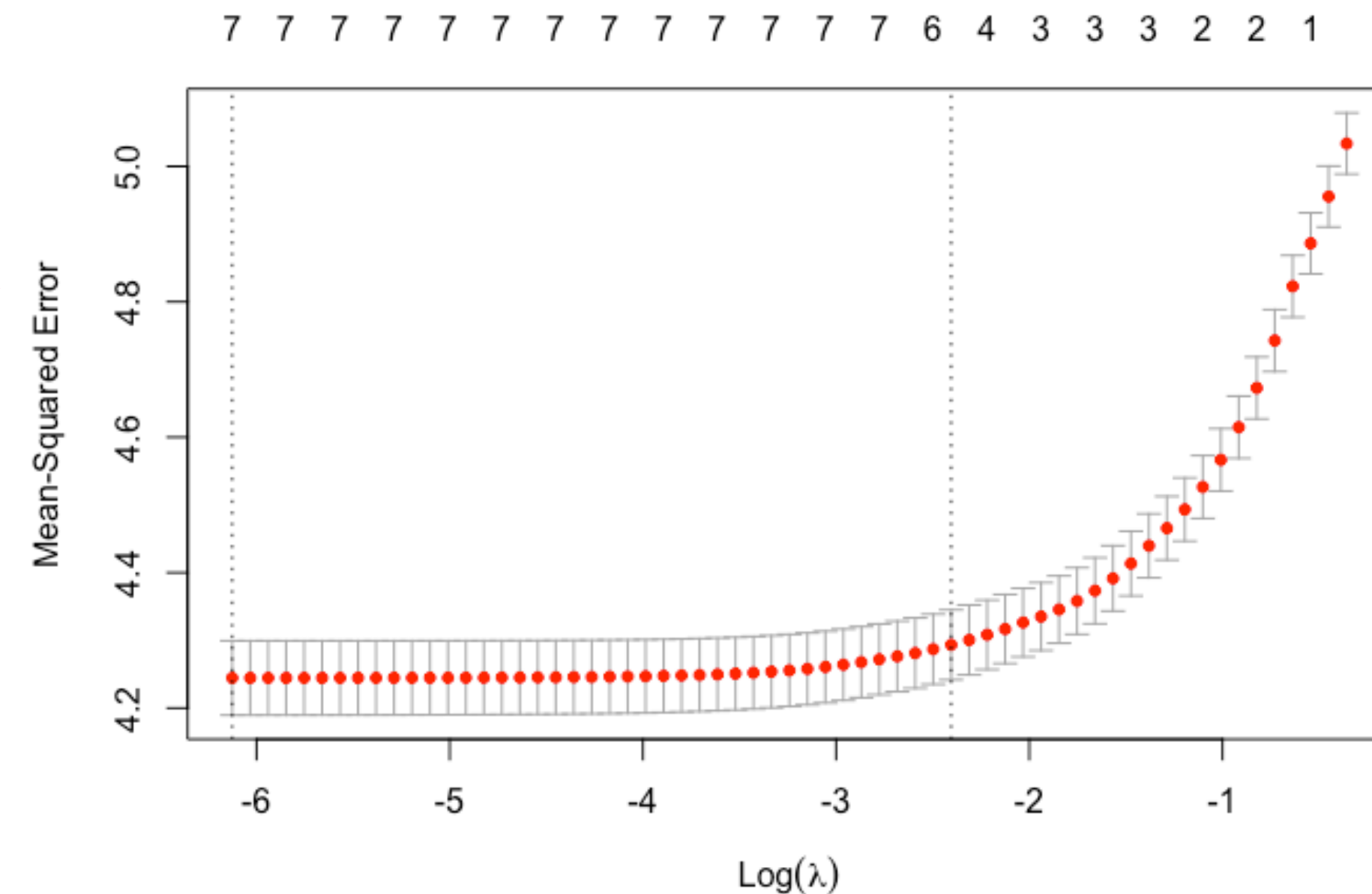
# Set the variables that are "fair game"
Y <- df_train$party_scale
X <- data.matrix(df_train[, c("age", "gender", "race_eth", "education",
                              "marital", "public_ins", "public_option")])
```

```
# LASSO regression
lasso <- cv.glmnet(x=X, y=Y)
plot(lasso); log(lasso$lambda.min)
```

```
[1] -6.125561
```

Here's all the $\log(\lambda)$ s plotted.

We want the minimum lambda, as this value minimizes the MSE.



Show me the predictions already!

Be sure to use the minimum lambda!



```
# In-sample prediction & MSE
```

```
df_train$predict_4 <- predict(lasso, newx=X, s="lambda.min")[, 1]  
df_train %>% summarise(mse = mean((party_scale - predict_4)^2))
```

```
mse
```

```
1 4.237769
```

```
# Get X variables of test set
```

```
X_test <- data.matrix(df_test[, c("age", "gender", "race_eth", "education",  
                                "marital", "public_ins", "public_option")])
```

```
# Out-of-sample prediction & MSE
```

```
df_test$predict_4 <- predict(lasso, newx=X_test, s="lambda.min")[, 1]  
df_test %>% summarise(mse = mean((party_scale - predict_4)^2))
```

```
mse
```

```
1 4.185614
```

Show me the predictions already!

```
# In-sample prediction & MSE
```

```
df_train$predict_4 <- predict(lasso, newx=X, s="lambda.min")[, 1]  
df_train %>% summarise(mse = mean((party_scale - predict_4)^2))
```

```
      mse  
1 4.237769
```



In-sample RMSE: $\sqrt{4.24} = 2.06$

```
# Get X variables of test set
```

```
X_test <- data.matrix(df_test[, c("age", "gender", "race_eth", "education",  
                                "marital", "public_ins", "public_option")])
```

```
# Out-of-sample prediction & MSE
```

```
df_test$predict_4 <- predict(lasso, newx=X_test, s="lambda.min")[, 1]  
df_test %>% summarise(mse = mean((party_scale - predict_4)^2))
```

```
      mse  
1 4.185614
```



Out-of-sample RMSE: $\sqrt{4.24} = 2.05$

So far, this is our best model! That said, performance will vary by context and available data.



LASSO

OLS

loves it when you can show it you're better than it
is at something it loves

Let's compare our models.

	In-sample RMSE	Out-of-sample RMSE
OLS (gender)	2.24	2.24
OLS (age + gender + race_eth + education)	2.12	2.13
OLS (age*gender*race_eth*education* marital*public_ins*public_option)	1.89	15.89
LASSO (age + gender + race_eth + education + marital + public_ins + public_option)	2.06	2.05

Of the models we tried, the
LASSO one performs best.

Is there a better way...?



Predictions, Part 1.

**Remember to keep your eye on the prize:
It's all about predictive accuracy.**

Be careful not to overfit the model.

**LASSO regression can help reduce
overfitting by penalizing extra terms.**

But there are even better ways...

